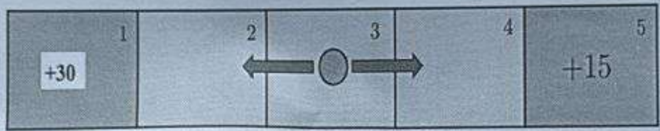
EE595H: Quiz
Total Questions: 2, Total Marks: 35, Time: 60 Mins

1. An absent-minded professor has three locations: Home, Office, and the Faculty Club. Each day, he commutes from Home $\rightarrow$ Office, then Office $\rightarrow$ Faculty Club, and finally Faculty Club $\rightarrow$ Home. He has a total of 3 umbrellas. If it rains (probability p) and he has an umbrella at his current location, he takes it. If it doesn't rain, he leaves all umbrellas behind. Rain events for each commute are independent.

   Find the steady-state probability that he gets wet on his commute from the Faculty Club back Home. $\boxed{10}$

2. Consider a simplified game problem defined over the grid shown below. The figure shows two terminal states $s \in \{1, 5\}$ with rewards $r(1) = +30, r(5) = +15$, and three additional states $s \in \{2, 3, 4\}$ constant reward $r(s) = -1$. The terminal squares lead to the EXIT state where the game stops. In the center states, the agent can select equally between two actions: $a \in$ {RIGHT, LEFT}. Assume that whenever the action RIGHT is chosen, there is a 25% chance that the agent will move left. There is no randomness with the action LEFT. The discount factor is set to $\gamma = 0.5$.



   (a) Determine the state transition probability matrix, P^{π}. $\boxed{5}$
   (b) Determine the one-step expected reward vector, r^{π}. $\boxed{5}$
   (c) Determine the state values, $v^{\pi}(s)$, under the assumed policy. $\boxed{5}$

$\boxed{\text{Total: 25}}$